

Submission Summary

Conference Name

4th International Conference on Advancements in Smart, Secure And Intelligent Computing

Paper ID

1352

Paper Title

NEXT GEN PRECISION FARMING USING ML FOR SUSTAINABLE AGRICULTURAL PRACTICES

Abstract

In light of current environmental changes, precision agriculture is critical to helping increase crop yield and promoting sustainable farming methods. Traditional farming methods often lack the capacity to adjust to changing conditions such as soil differences and variations in the weather leading to inefficient resource usage. This paper proposes a next-generation precision farming system that integrates IoT-based soil moisture sensors, real-time Weather API's and a TabNet deep learning model used for intelligent crop recommendations. By continuously monitoring the environmental parameters (soils moisture, air temperature, humidity, rainfall, and nutrient levels), the system will use this information to provide accurate and adaptive suggestions of crops to be planted. TabNet will provide high levels of prediction accuracy and will improve model interpretability by allowing features to be assigned attention. By using experimental data, we show that our model will achieve prediction accuracy up to 97.35%, an improvement over other traditional machine-learning methods such as Decision Tree and Random Forest. Additionally, the proposed system will enhance decision-making and will optimise resource use and promote sustainable farming practices.

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Submission Files

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